Dibaloke Chanda

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PROFESSIONAL SUMMARY

Ph.D. student in Computer Science specializing in **computational pathology, whole-slide image analysis, and multimodal machine learning**. Experienced in developing multiple-instance learning frameworks, pathology foundation models, vision-language systems, and representation-learning methods for cancer classification and biomedical image analysis. Strong background in PyTorch, WSI feature extraction, multimodal fusion, foundation-model evaluation, and scalable GPU-based experimentation.

SKILLS SUMMARY

- **Generative AI & Agentic Systems**: LLMs, multimodal LLMs/VLMs, RAG, GraphRAG, LangChain, LangGraph, CrewAI, vector databases
- **Self-Supervised Learning (SSL):** SimCLR, MoCo, BYOL, DINO, DINOv2, iBOT, MAE, I-JEPA, VICReg, Barlow Twins, SwAV
- **Machine Learning**: PyTorch, TensorFlow, JAX, PyTorch Geometric, Hugging Face Transformers, scikit-learn, OpenCV, Pandas, NumPy, SciPy, Seaborn
- **ML Systems & Infrastructure**: vLLM, SLURM, Docker, Linux, multi-GPU computing
- **Programming**: Python, C, C++, R, MATLAB, Java, JavaScript
- **Developer Tools:** Git, SQL, MCP
- **Computational Pathology:** WSI Image Processing, Feature Extraction, Multi-Instance Learning, Pathology Foundation Models, Vision-Language Model

EDUCATION

Ph.D. in Computer Science, GPA 4.00/4.00 **Jan. 2023 – May. 2027**
Marquette University, Milwaukee, WI

M.S. in Data Science — Specialization in Statistics and Modeling, GPA 4.00/4.00 **Jan. 2023 – Dec. 2024**
Marquette University, Milwaukee, WI

B.Sc. in Electrical, Electronic and Communication Engineering, CGPA 3.98/4.00 **Feb. 2017 – Feb. 2021**
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

RESEARCH AND TECHNICAL EXPERIENCE

Graduate Research Assistant **Jan. 2023 – Present**
Marquette University, Milwaukee, WI

- Developed **MERI-VL-MIL**, a multi-expert vision–language multiple-instance learning framework that integrates expert-guided reasoning with visual representations for few-shot whole-slide image classification; accepted to **IEEE ICASSP 2026**.
- Developed **MOSAIC**, a representation-learning framework for aligning **heterogeneous pathology foundation models** within a shared latent space, enabling cross-model comparison and transfer across **4,000+ whole-slide images** from multi-cohort datasets.
- Developed **PathoTME**, a tumor-microenvironment representation framework that analyzes frozen foundation-model features alongside tissue, cellular, and spatial descriptors to characterize **biologically meaningful structure** within learned visual representations.
- Developed **PGVL-Gym**, a reproducible benchmark for systematically evaluating prompt-guided vision-language models across few-shot multimodal classification tasks, with standardized prompting, inference, and evaluation pipelines for comparing model behavior across architectures and datasets.
- Processed **200+ TB** of whole-slide imaging data across **160+ cohorts** and **25+ pathology foundation models**, supporting large-scale feature extraction, representation learning, and comprehensive model benchmarking.

Graduate Teaching Assistant **Jan. 2023 – Present**
Marquette University, Milwaukee, WI

- Supported undergraduate instruction in four courses: Software Development, Object-Oriented Software Design, Hardware Systems and Data Structures through laboratories, office hours, and technical mentoring.

Lecturer **Mar. 2021 – Dec. 2022**
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- Taught undergraduate courses in Random Signals and Processes, Digital Signal Processing, Numerical Techniques, and Digital Communication; mentored five students on deep-learning research projects.

SELECTED PROJECTS

CThoughtPathNet: Expert-Guided Vision–Language Reasoning for Few-Shot Cancer Subtyping

- Developed CThoughtPathNet, a multimodal vision–language framework integrating multi-scale (10×/20×) pathology image representations, expert-guided LLM reasoning, and learned patch prototypes for few-shot whole-slide image classification.
- Designed a multi-agent reasoning pipeline with pathology-specific experts, leveraging attention-based feature fusion to align thought-driven text embeddings with visual patch representations, enhancing multimodal understanding and few-shot cancer subtyping.

INVEX: Label-Free Evaluation of Pathology Foundation Models through Invariance and Representation Quality | Project

- Developed **INVEX**, a label-free framework that ranks pathology foundation models by jointly quantifying representation invariance and discriminability, enabling reliable evaluation without downstream task labels.
- Evaluated **22 pathology foundation models** across **16 datasets** spanning **8 tissue sites** and **1.2M+ image patches**, demonstrating strong agreement with downstream performance while providing an efficient, architecture-agnostic benchmark for foundation model assessment.

MOSAIC: Towards a Universal Latent Space for Computational Pathology Foundation Models | Project

- Developed a unified latent-space framework to investigate and align representations across **18 pathology foundation models**, leveraging 4,465 whole-slide images (WSIs) from **TCGA** and **CPTAC**.
- Conducted extensive cross-model analyses and downstream evaluations across lung and breast **cancer subtyping, tumor staging, and molecular prediction tasks** to investigate representation consistency, transferability, and generalizability across pathology foundation models.

SELECTED PUBLICATIONS (7/17)

1. **Dibaloke Chanda**, Nasim Yahya Soltani and Masoud Ganji, **"CThoughtPathNet: Expert-Guided Vision–Language Reasoning for Few-Shot Cancer Subtyping in Whole-Slide Pathology,"** *Under Review at TPAMI*, 2027.
2. **Dibaloke Chanda** and Praveen Madiraju, "**INVEX: Label-Free Pathology Foundation Model Ranking via Invariance and Expressiveness**," *IEEE International Conference on Big Data (IEEE BigData), 2026*. (Accepted)
3. **Dibaloke Chanda** and Praveen Madiraju, "**MOSAIC: Towards a Universal Latent Space for Computational Pathology Foundation Models**," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2027*. (Accepted)
4. **Dibaloke Chanda**, Nasim Yahya Soltani and Masoud Ganji, **"MERI-VL-MIL: Multi-Expert Reasoning-Informed Vision-Language Multi-Instance Learning for Few-Shot Cancer Subtyping,"** *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2026.
5. **Dibaloke Chanda**, Milan Aryal, Masoud Ganji, Anil Parwani, and Nasim Yahya Soltani, **"A New Era in Computational Pathology: A Survey on Foundation and Vision-Language Models,"** *Journal of Pathology Informatics*, 2026.
6. **Dibaloke Chanda**, Saba Heidari Gheshlaghi and Nasim Yahya Soltani, **"Explainability-Based Adversarial Attack on Graphs Through Edge Perturbation,"** *Knowledge-Based Systems* (2025).
7. **Dibaloke Chanda** and Nasim Yahya Soltani, **"A Heterogeneous Graph-Based Multi-Task Learning for Fault Event Diagnosis in Smart Grid,"** *IEEE Transactions on Power Systems* (2025).

HONORS AND AWARDS

Computer Science Summer Research Fellowship (CSSRF) **Summer 2025, 2026**

- Awarded for conducting research on uncertainty-aware few-shot cancer subtyping in pathology.

NMDSI Student Research Scholars Award **Summer 2024**

- Awarded for conducting research on graph neural network applications in smart grid.

Top 10 Finalist, Three-Minute Thesis (3MT) Competition **Spring 2024**

- Presented on Explainability-Based Adversarial Attack on Graph Neural Networks; arranged by the Graduate School of Marquette University.

SERVICES AND VOLUNTEER WORK

- **Student Co-Chair,** research subcommittee of the Northwestern Mutual Data Science Institute.
- **Reviewer** for Knowledge and Information Systems; Scientific Reports; Peer-to-Peer Networking and Applications; Discover Artificial Intelligence; Electrical Engineering; International Journal of Machine Learning and Cybernetics, IEEE Transactions on Smart Grid and IEEE Transactions on Power Delivery.